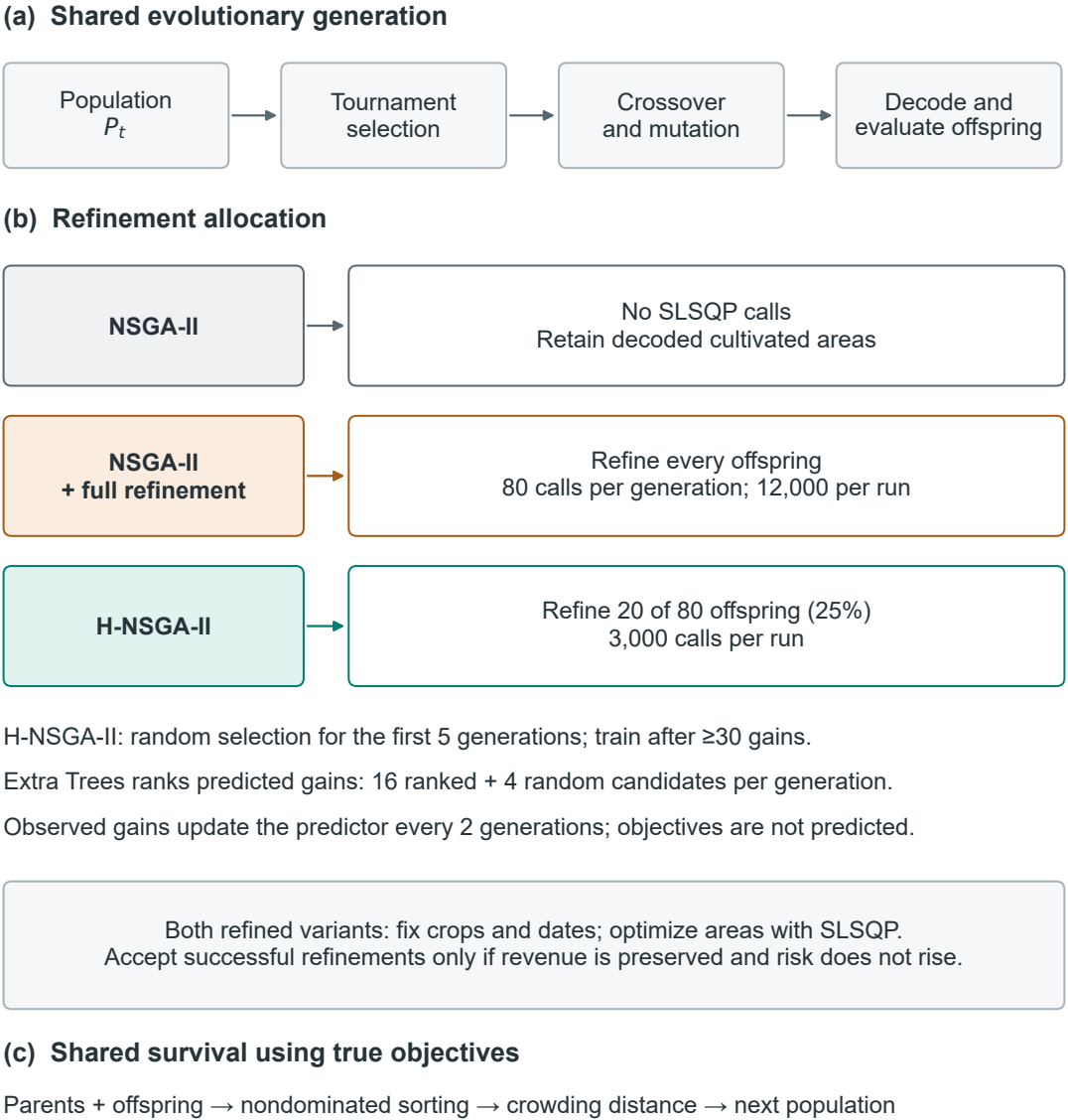

(a) Shared evolutionary generation
Population P_t
Tournament selection
Crossover and mutation
Decode and evaluate offspring
(b) Refinement allocation
NSGA-II
No SLSQP calls
Retain decoded cultivated areas
NSGA-II + full refinement
Refine every offspring
80 calls per generation; 12,000 per run
H-NSGA-II
Refine 20 of 80 offspring (25%)
3,000 calls per run
H-NSGA-II: random selection for the first 5 generations; train after ≥30 gains.
Extra Trees ranks predicted gains: 16 ranked + 4 random candidates per generation.
Observed gains update the predictor every 2 generations; objectives are not predicted.
Both refined variants: fix crops and dates; optimize areas with SLSQP.
Accept successful refinements only if revenue is preserved and risk does not rise.
(c) Shared survival using true objectives
Parents + offspring → nondominated sorting → crowding distance → next population